Result of Executed query in MYSQL

```
CREATE DATABASE mydb;

USE mydb;

-- 1. Create Customers Table

CREATE TABLE customers (

  customer_id INT PRIMARY KEY,

  customer_name VARCHAR(100) NOT NULL,

  region VARCHAR(50) NOT NULL,

  signup_date DATE NOT NULL,

  segment VARCHAR(50) DEFAULT 'Consumer'

);

-- 2. Create Orders Table

CREATE TABLE orders (

  order_id INT PRIMARY KEY,

  customer_id INT NOT NULL,

  order_date DATE NOT NULL,

  category VARCHAR(50) NOT NULL,

  product_name VARCHAR(100) NOT NULL,

  quantity INT NOT NULL,

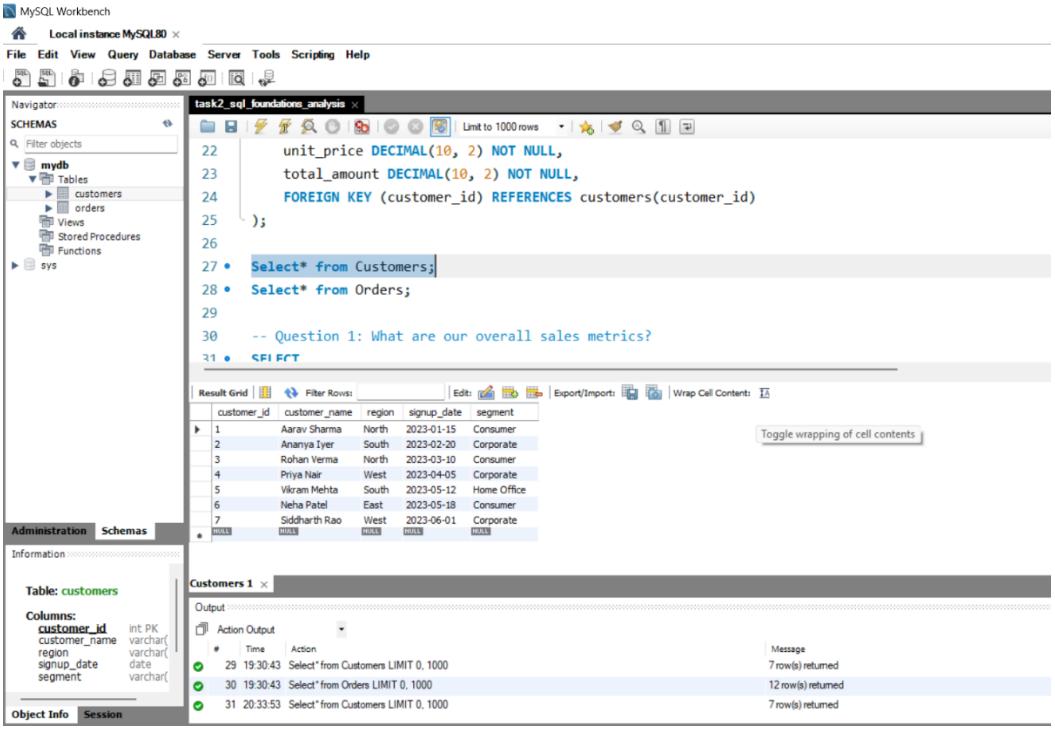
  unit_price DECIMAL(10, 2) NOT NULL,

  total_amount DECIMAL(10, 2) NOT NULL,

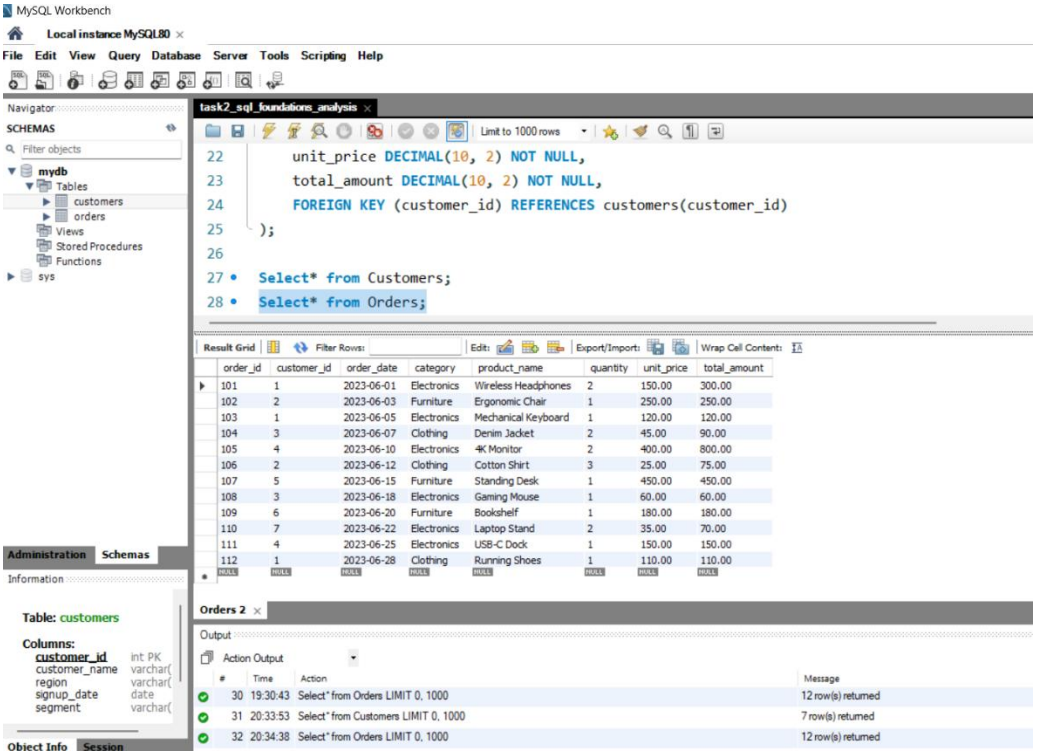
  FOREIGN KEY (customer_id) REFERENCES customers(customer_id)

);
```

```
Select* from Customers;
```

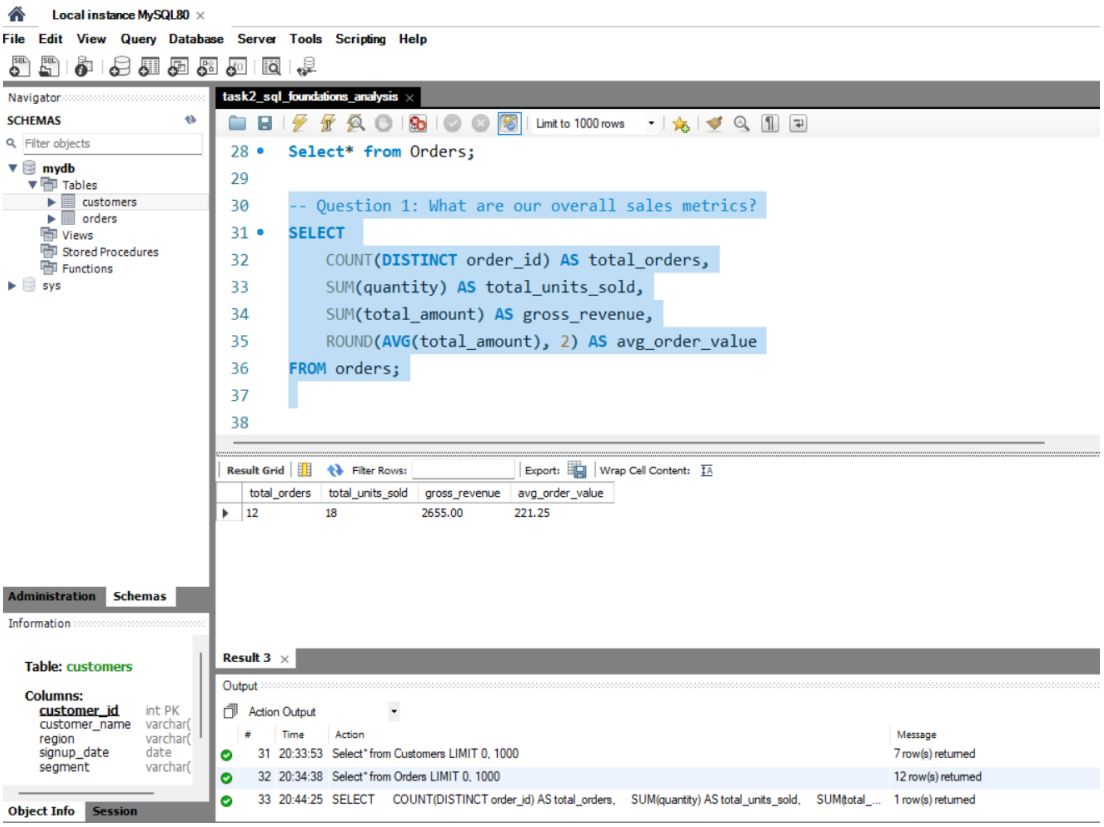


Select* from Orders;

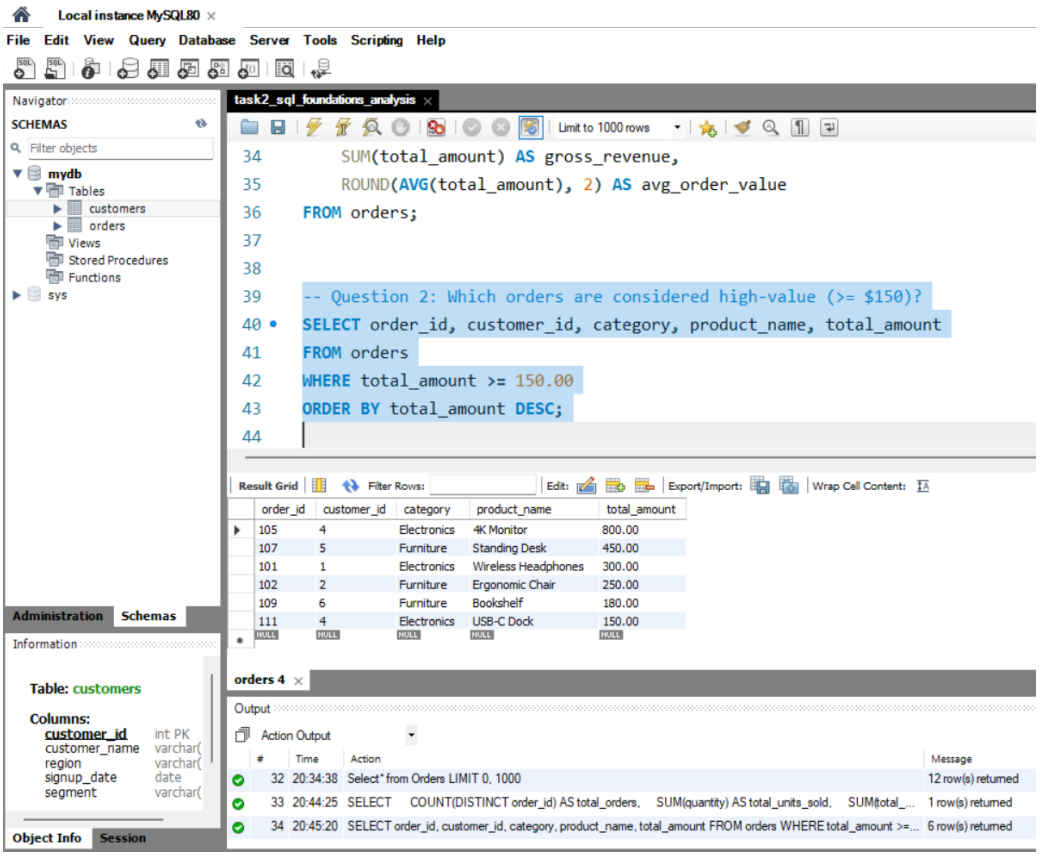


-- Question 1: What are our overall sales metrics?

```
SELECT
   COUNT(DISTINCT order_id) AS total_orders,
   SUM(quantity) AS total_units_sold,
   SUM(total_amount) AS gross_revenue,
   ROUND(AVG(total_amount), 2) AS avg_order_value
FROM orders;
```




```
-- Question 2: Which orders are considered high-value (>= $150)?
SELECT order_id, customer_id, category, product_name, total_amount
FROM orders
WHERE total_amount >= 150.00
ORDER BY total_amount DESC;
```



-- Question 3: What is our revenue breakdown by product category?

SELECT

category,

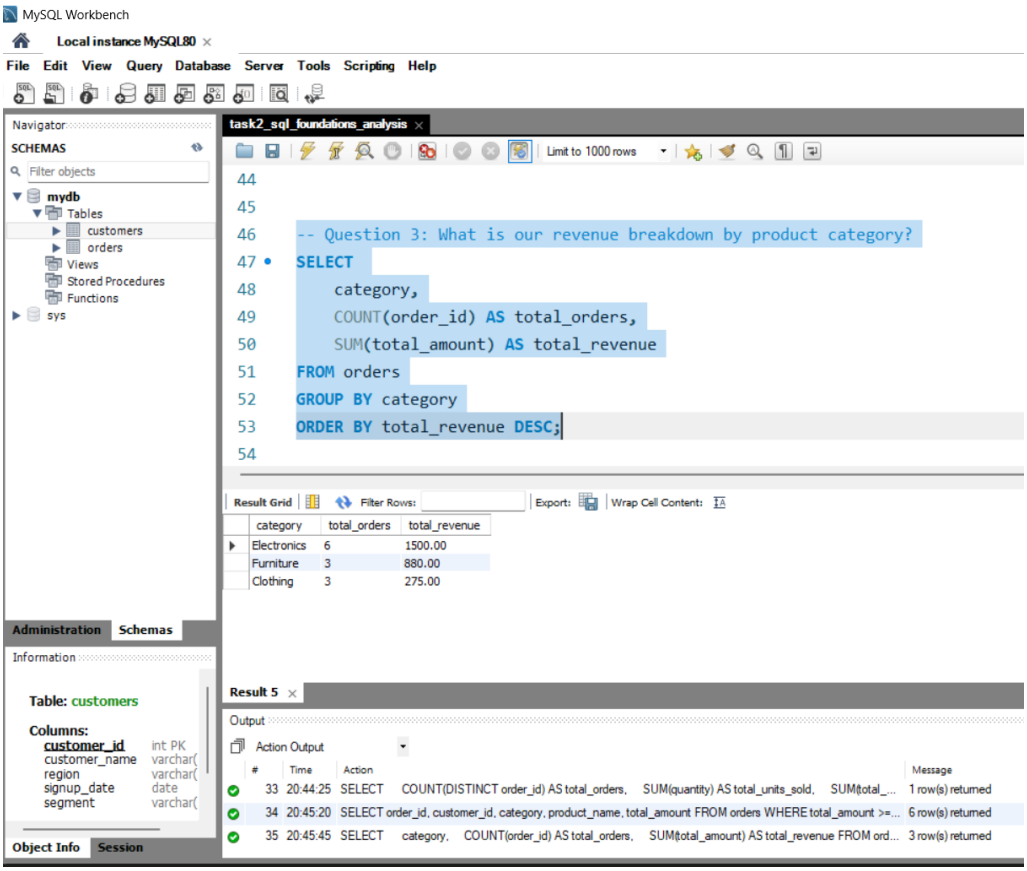
COUNT(order_id) AS total_orders,

SUM(total_amount) AS total_revenue

FROM orders

GROUP BY category

ORDER BY total_revenue DESC;



-- Question 4: What is the average order spend per category?

SELECT

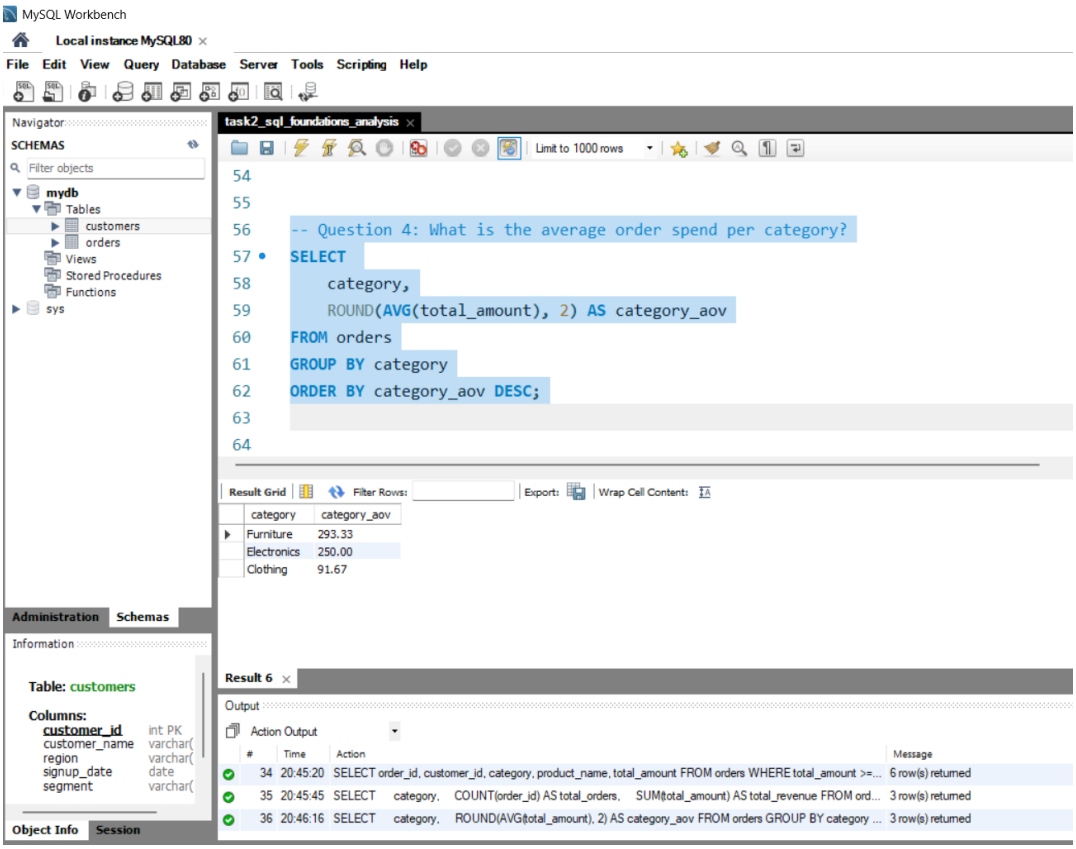
category,

ROUND(AVG(total_amount), 2) AS category_aov

FROM orders

GROUP BY category

ORDER BY category_aov DESC;



-- Question 5: How are our customers distributed geographically?

SELECT

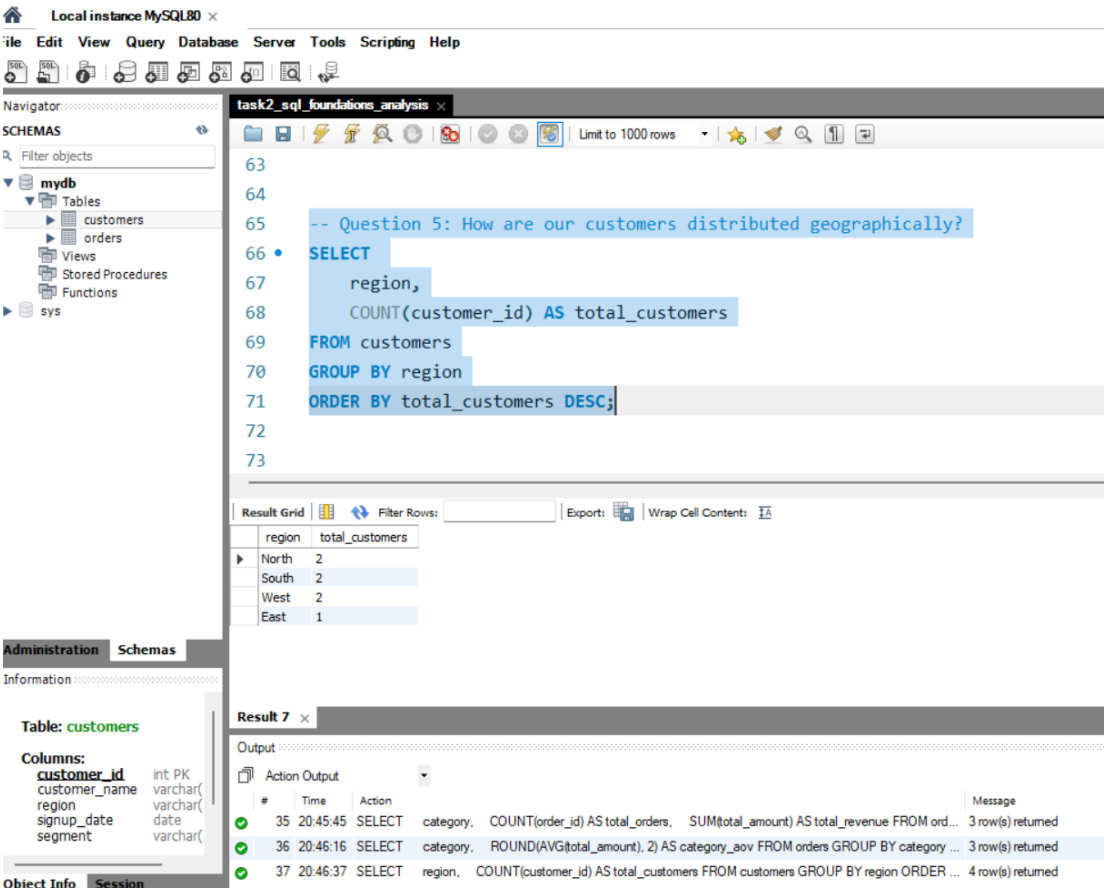
region,

COUNT(customer_id) AS total_customers

FROM customers

GROUP BY region

ORDER BY total_customers DESC;



-- Question 6: Who are our top customers by Lifetime Value (LTV)?

SELECT

c.customer_id,

c.customer_name,

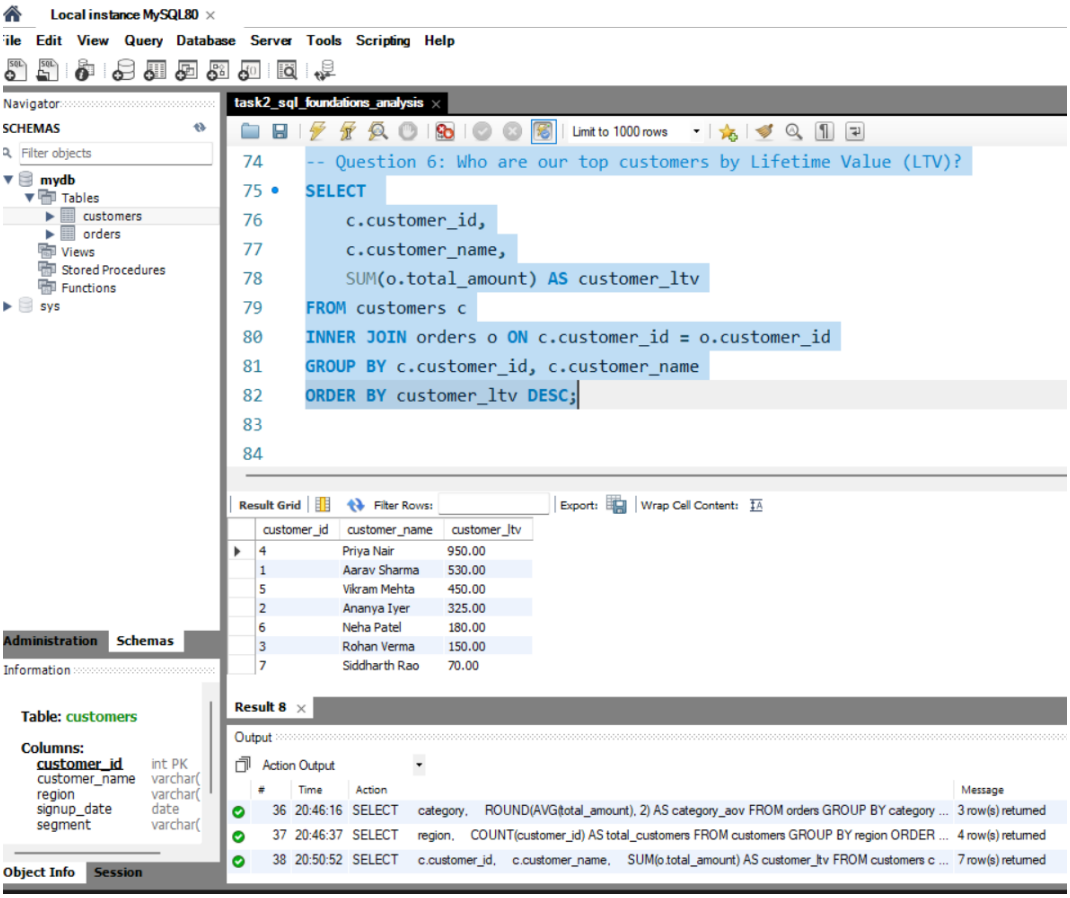
SUM(o.total_amount) AS customer_ltv

FROM customers c

INNER JOIN orders o ON c.customer_id = o.customer_id

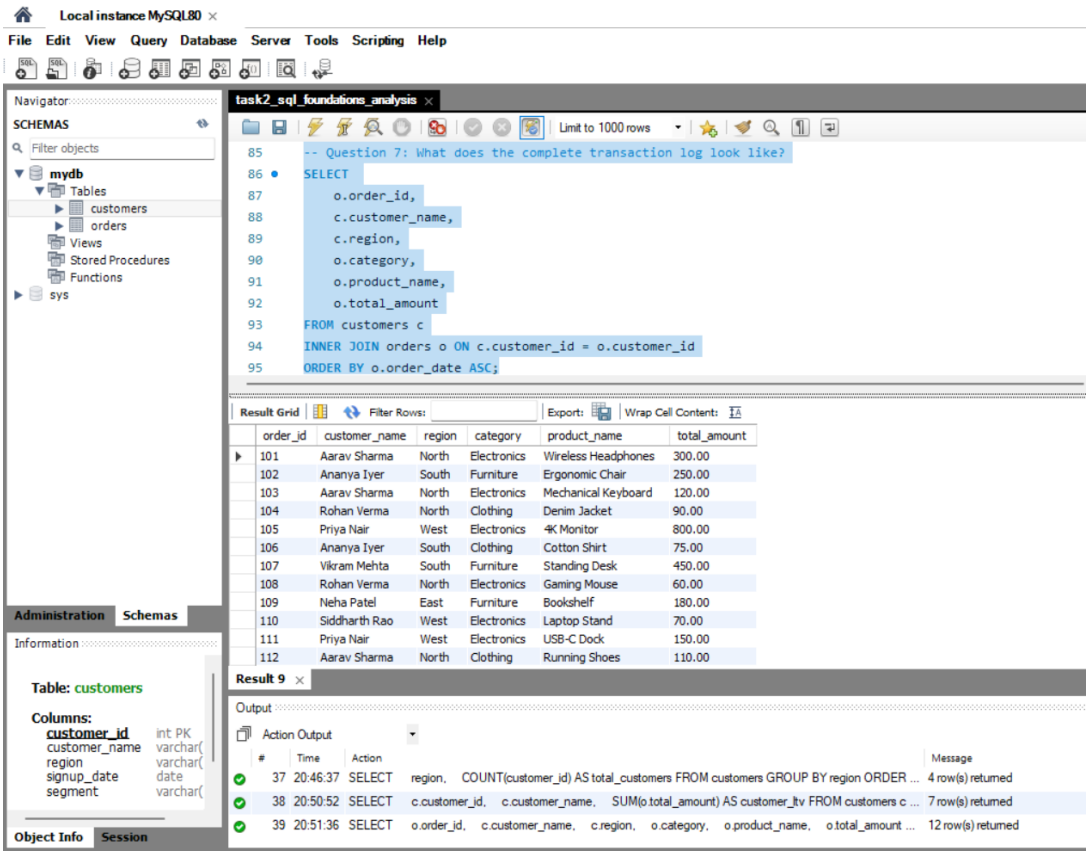
GROUP BY c.customer_id, c.customer_name

ORDER BY customer_ltv DESC;



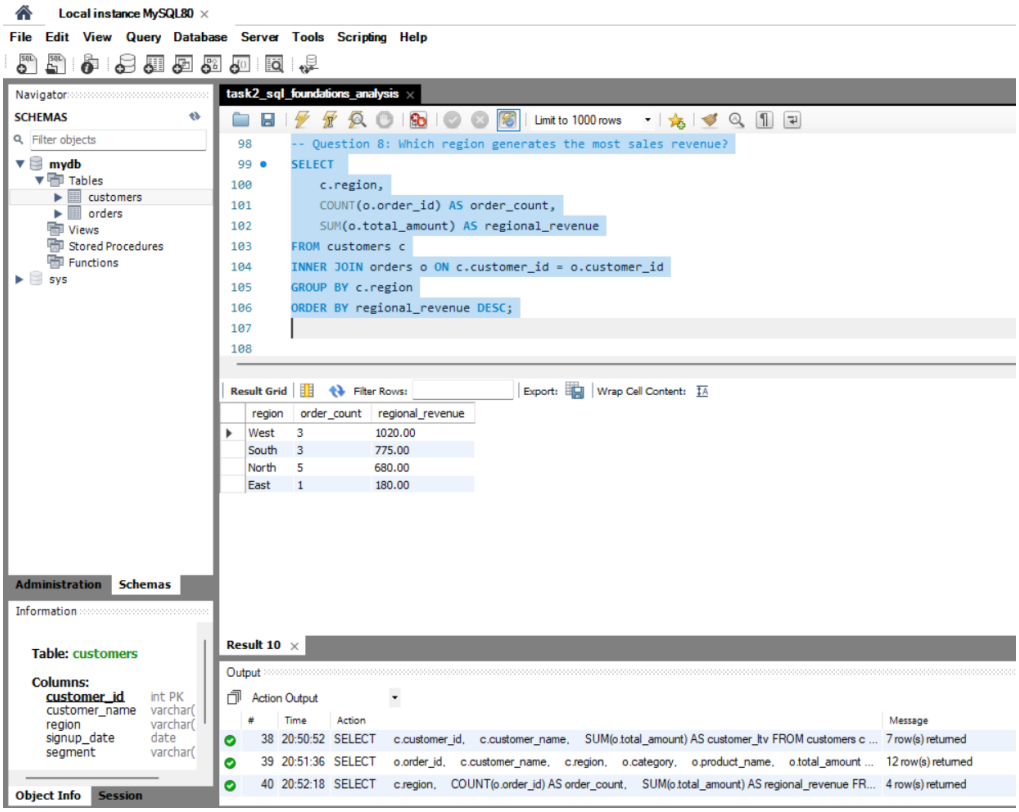
-- Question 7: What does the complete transaction log look like?

```
SELECT
    o.order_id,
    c.customer_name,
    c.region,
    o.category,
    o.product_name,
    o.total_amount
FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
ORDER BY o.order_date ASC;
```

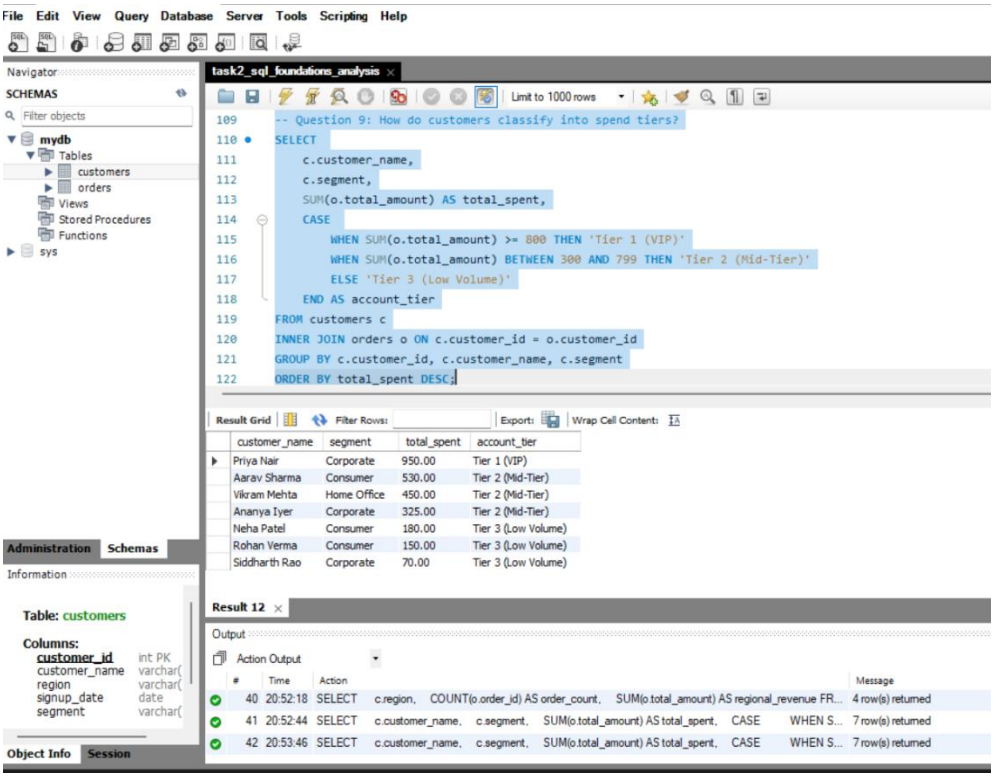


-- Question 8: Which region generates the most sales revenue?

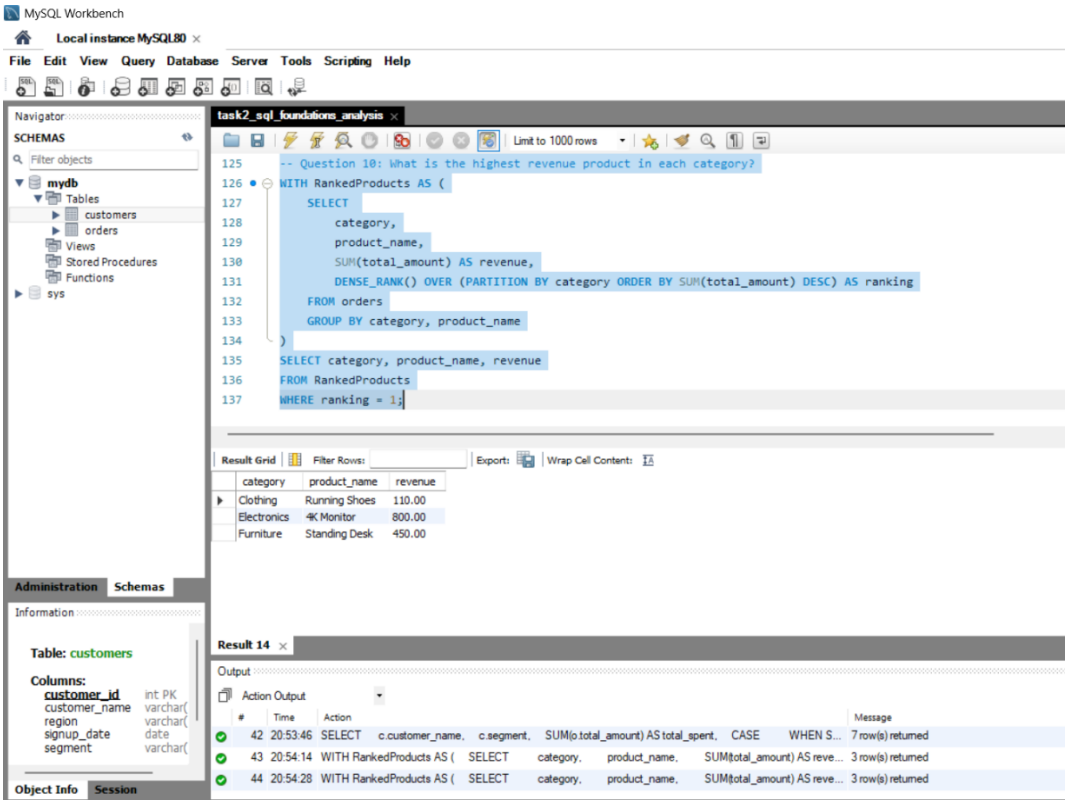
```
SELECT
   c.region,
   COUNT(o.order_id) AS order_count,
   SUM(o.total_amount) AS regional_revenue
FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.region
ORDER BY regional_revenue DESC;
```




```
-- Question 9: How do customers classify into spend tiers?
SELECT
  c.customer_name,
  c.segment,
  SUM(o.total_amount) AS total_spent,
  CASE
    WHEN SUM(o.total_amount) >= 800 THEN 'Tier 1 (VIP)'
    WHEN SUM(o.total_amount) BETWEEN 300 AND 799 THEN 'Tier 2 (Mid-Tier)'
    ELSE 'Tier 3 (Low Volume)'
  END AS account_tier
FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name, c.segment
ORDER BY total_spent DESC;
```




```
-- Question 10: What is the highest revenue product in each category?
WITH RankedProducts AS (
  SELECT
    category,
    product_name,
    SUM(total_amount) AS revenue,
    DENSE_RANK() OVER (PARTITION BY category ORDER BY SUM(total_amount) DESC) AS ranking
  FROM orders
  GROUP BY category, product_name
)
SELECT category, product_name, revenue
FROM RankedProducts
WHERE ranking = 1;
```


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
mydb
Tables
customers
orders
Views
Stored Procedures
Functions
sys
Administration Schemas
Information
Table: customers
Columns:
customer_id int PK
customer_name varchar(
region varchar(
signup_date date
segment varchar(
Object Info Session
task2_sql_foundations_analysis
Limit to 1000 rows
125 -- Question 10: What is the highest revenue product in each category?
126 WITH RankedProducts AS (
127 SELECT
128 category,
129 product_name,
130 SUM(total_amount) AS revenue,
131 DENSE_RANK() OVER (PARTITION BY category ORDER BY SUM(total_amount) DESC) AS ranking
132 FROM orders
133 GROUP BY category, product_name
134)
135 SELECT category, product_name, revenue
136 FROM RankedProducts
137 WHERE ranking = 1;
Result Grid Filter Rows: Export: Wrap Cell Content:
category product_name revenue
Clothing Running Shoes 110.00
Electronics 4K Monitor 800.00
Furniture Standing Desk 450.00
Result 14
Output
Action Output
Time Action Message
42 20:53:46 SELECT c.customer_name, c.segment, SUM(o.total_amount) AS total_spent, CASE WHEN S... 7 row(s) returned
43 20:54:14 WITH RankedProducts AS (SELECT category, product_name, SUM(total_amount) AS reve... 3 row(s) returned
44 20:54:28 WITH RankedProducts AS (SELECT category, product_name, SUM(total_amount) AS reve... 3 row(s) returned